UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 16

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Instructor: Somayeh Sojoudi

Typeset by: George Ma

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Convex optimization can be written as:

$$\begin{aligned} \min \quad & f_0(\boldsymbol{x}), \\ \text{s.t.} \quad & \boldsymbol{x} \in \mathcal{S}. \end{aligned}$$

Assume $f_0(\boldsymbol{x})$ is defined on $\mathcal{S}$, i.e., $\mathcal{S} \subseteq \operatorname{dom} f_0$. Otherwise we should write:

$$\begin{aligned} \min \quad & f_0(\boldsymbol{x}), \\ \text{s.t.} \quad & \boldsymbol{x} \in \mathcal{S} \cap \operatorname{dom} f_0. \end{aligned}$$

Theorem 1. *For a convex optimization:*

(i) Any local minimum $\boldsymbol{x}^*$ is a global minimum.

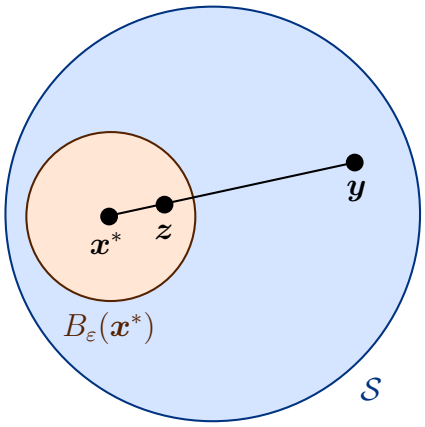
(ii) Let $\mathcal{X}^{\text{opt}}$ denote the set of all global minima, then $\mathcal{X}^{\text{opt}}$ is a convex set.

Proof. We only prove (i). Check the textbook for proof of (ii).
Let $\boldsymbol{x}^*$ denote a local minimum. By the local optimality of $\boldsymbol{x}^*$, there exists a ball $B_\varepsilon(\boldsymbol{x}^*)$ around $\boldsymbol{x}^*$ such that:

$$f(\boldsymbol{x}^*) \le f(\boldsymbol{x}), \quad \forall \boldsymbol{x} \in B_\varepsilon(\boldsymbol{x}^*).$$

Consider an arbitrary point $\boldsymbol{y} \in \mathcal{S}$. Connect $\boldsymbol{y}$ to $\boldsymbol{x}^*$ to form a segment. Since $\mathcal{S}$ is convex, the segment is in $\mathcal{S}$. We pick a point $\boldsymbol{z}$ on the segment and inside the ball $B_\varepsilon(\boldsymbol{x}^*)$:

$$\boldsymbol{z} = \alpha \boldsymbol{x}^* + (1-\alpha)\boldsymbol{y}, \quad \text{for some } 0 < \alpha < 1.$$



Since $\boldsymbol{z} \in B_\varepsilon(\boldsymbol{x}^*)$, we have

$$f(\boldsymbol{x}^*) \le f(\boldsymbol{z}) = f\big(\alpha \boldsymbol{x}^* + (1-\alpha)\boldsymbol{y}\big).$$

By the convexity of f,

$$f(\boldsymbol{x}^*) \le f\big(\alpha \boldsymbol{x}^* + (1-\alpha)\boldsymbol{y}\big) \le \alpha f(\boldsymbol{x}^*) + (1-\alpha) f(\boldsymbol{y}).$$

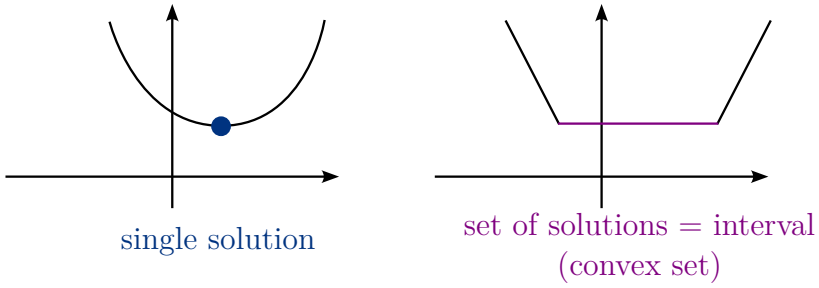
Rearrange, we get

$$(1-\alpha) f(\boldsymbol{x}^*) \le (1-\alpha) f(\boldsymbol{y}).$$

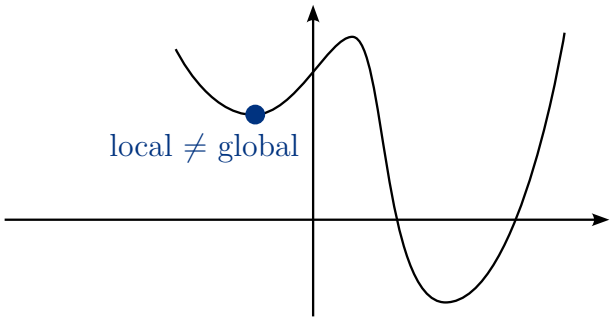
Since $\alpha < 1$, this gives $f(\boldsymbol{x}^*) \le f(\boldsymbol{y})$. By the arbitrariness of $\boldsymbol{y}$, $\boldsymbol{x}^*$ is a global minimum. ■

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Example 1. Consider $\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x})$ where $\nabla^2 f(\boldsymbol{x}) \succeq \mathbf{0}$, $\forall \boldsymbol{x} \in \mathbb{R}^n$. Then, $f(\boldsymbol{x})$ is a convex function, and by Theorem 1, every local minimum is a global minimum.
This corresponds to:



For problems like:



it is not convex optimization. ■

Example 2. Consider

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^2} \quad & (x_1 - x_2)^6 + 10(x_1 + 5x_2), \\ \text{s.t.} \quad & \mathrm{e}^{x_1+10x_2} + 2\mathrm{e}^{x_1-x_2} \leq -x_1 + 2. \end{aligned}$$

This is a convex optimization problem. By Theorem 1, any local minimum is also a global minimum. ■

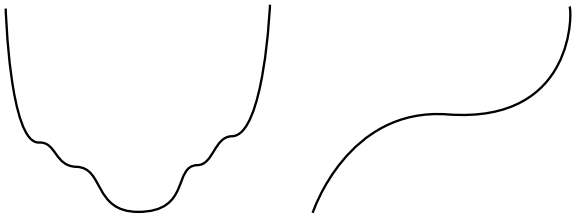
Consider $\min \mathrm{e}^x$. This is convex optimization, but has no solution ($\boldsymbol{x}^* = -\infty$ is not a number).
How to guarantee that the solution exists (i.e., it is finite)?
We need the *coercive* notion:

Definition 1 (Coercive function)**.** A function $f\colon \mathbb{R}^n \to \mathbb{R}$ (convex or non-convex) is called *coercive* if $\lim_{\|\boldsymbol{x}\|\to\infty} f(\boldsymbol{x}) = +\infty$.

This means when $\boldsymbol{x}$ goes to infinity in any possible direction (i.e., $\|\boldsymbol{x}\| \to \infty$), $f(\boldsymbol{x})$ always blows up to $+\infty$.

Example 3 (Coercive function for $n = 1$)**.** For $n = 1$, $\|x\| \to \infty$ means $x \to +\infty$ or $x \to -\infty$. Thus, a coercive function satisfies $f(+\infty) = +\infty$ and $f(-\infty) = +\infty$.

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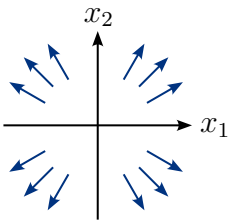
coercive non-coercive

$f(x) = \mathrm{e}^x$ is not coercive since $\mathrm{e}^{-\infty} \neq +\infty$. ■

Example 4 (Coercive function for $n = 2$)**.** For $n = 2$, $\|\boldsymbol{x}\| \to \infty$ means

$$\boldsymbol{x} \to \begin{bmatrix} \pm\infty \\ \text{arbitrary} \end{bmatrix} \quad \text{or} \quad \boldsymbol{x} \to \begin{bmatrix} \text{arbitrary} \\ \pm\infty \end{bmatrix},$$

and $f(\boldsymbol{x})$ should grow towards $+\infty$.



Blue arrow means $\|\boldsymbol{x}\| \to \infty$
Then $f(\boldsymbol{x})$ drawn in 3-dim should blow up

■

Example 5 (Coercive function)**.** $f(\boldsymbol{x}) = \alpha_1 x_1^4 + \alpha_2 x_2^4$ is coercive for all $\alpha_1, \alpha_2 > 0$.

Proof. We have

$$\begin{aligned} f(\boldsymbol{x}) &\geq \min(\alpha_1, \alpha_2) \cdot (x_1^4 + x_2^4) \\ &\geq \min(\alpha_1, \alpha_2) \cdot \frac{(x_1^2 + x_2^2)^2}{2} \\ &= \min(\alpha_1, \alpha_2) \cdot \frac{\|\boldsymbol{x}\|^4}{2}. \end{aligned}$$

The right-hand side goes to $+\infty$ if $\|\boldsymbol{x}\| \to \infty$. ■

If $\alpha_1 = 0$, it is not coercive since $f\left(\begin{bmatrix} x_1 \\ 0 \end{bmatrix}\right) = 0$, and as $x_1 \to \infty$, f does not grow to $+\infty$.

If $\alpha_1 < 0$, then as $x_1 \to \infty$, $f\left(\begin{bmatrix} x_1 \\ 0 \end{bmatrix}\right) \to -\infty$, so it is again not coercive. ■

Theorem 2. *Consider $f\colon \mathbb{R}^n \to \mathbb{R}$ (convex or non-convex) with $\operatorname{dom} f = \mathbb{R}^n$. Assume f is continuous and coercive. Then, the optimization $\min f(\boldsymbol{x})$ has a finite solution.*

Here $\min f(\boldsymbol{x})$ is unconstrained optimization, but we can generalize it to constrained optimization:

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Theorem 3.

(i) Consider $\min f(\boldsymbol{x})$ *s.t.* $\boldsymbol{x} \in \mathcal{S}$ *where* f *(convex or non-convex) is coercive and continuous with* $\operatorname{dom} f = \mathbb{R}^n$*. If* $\mathcal{S}$ *(convex or non-convex) is closed and non-empty, then the optimization has a finite solution.*

(ii) Consider

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f_0(\boldsymbol{x}), \\ s.t. \quad & f_i(\boldsymbol{x}) \le 0,\ i = 1, \dots, m, \\ & h_j(\boldsymbol{x}) = 0,\ j = 1, \dots, k, \end{aligned}$$

where f_0*,* f_i*'s,* h_j*'s are arbitrary but continuous with domain* $\mathbb{R}^n$*, and* f_0 *is coercive. If the feasible set is non-empty, then the optimization has a finite solution.*

- Note for (i): If $\mathcal{S}$ is not closed, it could be a problem. For example, consider $\min \mathrm{e}^x$ s.t. $x \in \mathcal{S}$ with $\mathcal{S} = (0, 1]$. The optimal objective value is e^0, but 0 is not in $\mathcal{S}$ so there is no solution.
- Note for (ii): If the domain is not $\mathbb{R}^n$, it could be a problem. For example, consider $\min \log x$ s.t. $x \ge 0$, where $\operatorname{dom} \log x = (0, +\infty)$. Since $\log 0 = -\infty$ and $\log x$ is not defined at $x = 0$, there is no solution.

What if $f(\boldsymbol{x})$ is not coercive? There is an alternative theorem:

Theorem 4 (Weierstrass theorem)**.** *Consider* $\min f(\boldsymbol{x})$ *s.t.* $\boldsymbol{x} \in \mathcal{S}$*, where* $f \colon \mathbb{R}^n \to \mathbb{R}$ *is continuous. If* $\mathcal{S}$ *is compact (i.e., closed and bounded) and non-empty, then the optimization has a finite solution.*

In terms of general optimization:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f_0(\boldsymbol{x}), \\ \text{s.t.} \quad & f_i(\boldsymbol{x}) \le 0,\ i = 1, \dots, m, \\ & h_j(\boldsymbol{x}) = 0,\ j = 1, \dots, k. \end{aligned}$$

If f_i's and h_j's are continuous, then the feasible set is closed. So, we only need the feasible set to be bounded and f_0 to be continuous to guarantee we have a finite solution.

Example 6. Consider

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^2} \quad & \mathrm{e}^{x_1} - x_1^2 + x_2^2 + \mathrm{e}^{x_1+x_2}, \\ \text{s.t.} \quad & \cos x_1 + \sin x_2 \le 1, \\ & x_1^4 + \mathrm{e}^{x_2^2} \le 5. \end{aligned}$$

Observe that:

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(1) All functions in this optimization are defined over $\mathbb{R}^2$, so the domain is the entire space.

(2) The constraint $x_1^4 + e^{x_2^2} \leq 5$ makes the feasible set bounded, since if either x_1 or x_2 goes to $\pm\infty$, this constraint wouldn't be met.

Therefore, by Theorem 4, the solution exists.
Note: If $e^{x_2^2}$ is changed to e^{x_2}, it would be a problem since $x_1^4 + e^{x_2} \leq 5$ is satisfied for $x_1 = 0$ and $x_2 = -\infty$, and the feasible set could be unbounded in this case. ■

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